# Mayukh Haldar

Kolkata, West Bengal, India | +91-9432182153
LinkedIn | mayukhhaldar1@gmail.com
GitHub


## Skills

- **Programming:** ***Python (NumPy, Pandas, Matplotlib, Seaborn), C, C++, SQL***, *Java*
- **Machine Learning & Deep Learning:** ***Scikit-learn, TensorFlow, PyTorch***, *Multi-label Classification,* ***Model Evaluation, Feature Engineering***
- **Data Analysis:** ***EDA, Data Cleaning, Statistical Analysis***, *Dimensionality Reduction (**PCA, UMAP**)*
- **Research & LLMs:** ***LLM Finetuning*** *(Unsloth AI),* ***Prompt Engineering, Hugging Face, AutoTrain, LLaMA models***
- **Generative AI & Diffusion Models:** ***Stable Diffusion, LoRA Training*** *(ComfyUI, Kohya),* ***A1111 Inference***
- **Databases & Tools:** ***MySQL, MongoDB***, *Excel (Pivot Tables),* ***Power BI, Tableau***

## Projects

- **Final Year Project – MoA Cell Viability Analysis (EDA + Model Training)** *2025*

  *Comprehensive project involving* ***Exploratory Data Analysis (EDA)*** *and* ***Deep Learning*** *model development for* ***Mechanisms of Action (MoA)*** *prediction. Built and evaluated architectures including* ***Multi-Stage Neural Networks, 1D-CNNs, DeepInsight ResNeSt,*** *and* ***blended models****. Achieved strong leaderboard performance with optimized* ***training pipelines*** *and* ***feature engineering****.*
  - **Achievements:** Kaggle Private LB: **0.01645**, Public LB: **0.01868**
  - **Performed: EDA**, **PCA/UMAP** visualization, **feature correlation** studies, **drug-wise stratification**, **model comparison**
  - **Models: Multi-stage NNs**, **1D-CNN**, **DeepInsight CNNs**, **geometric logit blending**
  - **Outcome:** Built a high-performance **MoA prediction pipeline** aligned with **SOTA Kaggle solutions**

- **Kaggle LLM Prompt Recovery** *2024*

  *Participated in Kaggle's* ***LLM Prompt Recovery*** *competition, focused on reconstructing prompts used by* ***Gemma****, Google's open LLM. Built complete* ***NLP pipelines*** *using* ***keras_nlp, NumPy, Pandas****, and visualization tools. Achieved strong leaderboard results.* View Notebook
  - **Scores:** Public LB: **0.5182**, Private LB: **0.5195**
  - **Performed: Prompt design**, **dataset preprocessing**, **evaluation metric optimization**
  - **Tech Stack: keras_nlp**, **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**

- **CyberPunk Style LoRA Model – Civitai** *2024*

  *Developed and released a* ***specialized LoRA model*** *called* ***CyberPunk Style LoRA*** *using* ***Stable Diffusion 1.5*** *and* ***DreamShaper 8****. Created a curated* ***cyberpunk-themed dataset*** *and built a complete* ***training + inference pipeline****. Published on* ***Civitai****.* View Model
  - **Tools: WD 1.4 Tagger**, **ComfyUI** (LoRA training + inference), **A1111** for testing
  - **Outcome:** Produced a **high-quality cyberpunk-style model** widely used by the creator community
  - **Skills: Data preprocessing**, **model training**, **prompt engineering**, **deployment**

## Experience

- **Associate Analyst – PDI CMS Intern** *Jan 2026 – Present*

  **Deloitte India**
  - Selected as **Associate Analyst** for the **Platform Development & Integration (PDI)** and **Cloud Managed Services (CMS)** track
  - Currently completing a structured **capstone project** prior to corporate internship onboarding
  - Working on enterprise-grade **cloud platform integration**, **managed services workflows**, and deployment best practices
  - Gaining exposure to **cloud infrastructure, system monitoring, reliability engineering, and integration architecture**

- **Data Science & Machine Learning Intern** *Aug 2024 – Oct 2024*

  **WEBEL (West Bengal Electronics Industry Development Corporation Limited)**
  - Designed and developed **Proshno Sathi**, an AI-powered multilingual platform to search and discover **government schemes**
  - Integrated **Speech-to-Text (STT)**, **real-time translation**, and **LLM-based response generation** for improved accessibility
  - Built an end-to-end **NLP pipeline** with real-time query processing and multilingual support
  - Developed responsive **frontend components** to enhance user experience (UX)
  - Deployed the system using **Render** with production-ready architecture

## Honors & Achievements

- **ICPC Amritapuri Prelims 2025-26:** Secured **Institute Rank 1** and **All India Rank 1461**.
  Solved **3 out of 6 problems**. ICPC ID Link | Leaderboard Link *2025*
- **Meta Hacker Cup 2025:** Qualified for **Round 2** by solving **3 out of 6 problems** in Round 1. Verification Link *2025*

## Education

- **RCC Institute of Information Technology** *Oct 2022 - July 2026*

  *Bachelor of Technology (B.Tech.) - Computer Science and Engineering* **CGPA: 8.77 (upto 6th Sem)**
- **Bhavan's Gangabux Kanoria Vidyamandir, Kolkata** *Aug 2020 - Jun 2022*

  *All India Senior School Certificate Examination, CBSE* **Percentage: 96.4 (Best of 5)**
- **Bhavan's Gangabux Kanoria Vidyamandir, Kolkata** *Jun 2009 - Jun 2020*

  *All India Secondary School Examination, CBSE* **Percentage: 96.4 (Top 5)**

## Certifications

- **CodinGame Python 3 Certification** Verify *2023*
- **HackerRank Python (Basic) Certificate** Verify *2023*